

Differential Equations for Master Integrals

A concise overview

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These notes summarize a longer treatment. The setting is the double-real channel of $pp \rightarrow h + X$ at NNLO, in $d = 4 - 2\varepsilon$ dimensions. Reverse unitarity replaces each on-shell delta function by the discontinuity of a propagator, on positive-energy configurations,

$$\delta^{(+)}(p^2) = \frac{1}{2\pi i} \left[\frac{1}{p^2 - i0} - \frac{1}{p^2 + i0} \right], \quad (1)$$

so that the phase-space integrals become cut two-loop integrals: they obey the ordinary integration-by-parts identities, a cut line raised to power a_i standing for $[(-1)^{a_i-1}/(a_i-1)!] \delta^{(a_i-1)}$ and an integral with a cancelled cut line vanishing. Reduction leaves 347 master integrals. A master carries the overall scale by its mass dimension, $I_m = (-s)^{Ld/2-a_m} \hat{I}_m$ with L loops and propagator powers summing to a_m , and the equations below govern the dimensionless $\hat{I}_m$. For the partonic process $p_1 + p_2 \rightarrow p_3 + k_1 + k_2$ with all partons massless, and $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, $u = (p_2 - p_3)^2$, these depend on the two ratios $v = -t/s$ and $w = -u/s$ alone. Since $s+t+u = (p_1+p_2-p_3)^2 = M^2$ is the squared invariant mass of the unobserved pair, $1 - v - w = M^2/s \geq 0$, while $v, w > 0$: the physical region is the triangle $0 < v$, $0 < w$, $v + w < 1$. In the partonic centre-of-mass frame $v = (E_3/\sqrt{s})(1 - \cos\theta)$ and $w = (E_3/\sqrt{s})(1 + \cos\theta)$, so $v \rightarrow 0$ and $w \rightarrow 0$ are the collinear limits and $v + w \rightarrow 1$ the Born configuration $E_3 = \sqrt{s}/2$, $M^2 = 0$, at which the radiation is unresolved: the soft edge. The problem is the analytic evaluation of the masters: each as a Laurent series in ε whose coefficients are explicit transcendental functions with exact constants.

1 Differential equations

A master depends on the external momenta only through their invariants s_{kl} , so the chain rule $p_j^\mu \partial_{p_i^\mu} I = \sum_{k \leq l} (p_j^\mu \partial s_{kl} / \partial p_i^\mu) \partial I / \partial s_{kl}$ is a linear system for the $\partial I / \partial s_{kl}$, invertible wherever the Gram determinant of the external momenta is non-zero; inverting it expresses ∂_v and ∂_w through the operators $p_j^\mu \partial / \partial p_i^\mu$. Taken under the integral sign these raise propagator powers and produce numerators which, the family being complete, are linear combinations of its inverse propagators and of invariants; a derivative of a master is therefore again a combination of integrals of the same family, and integration by parts brings it back to the masters of the family and its subsectors. Each family therefore satisfies a closed first-order system in both variables at once,

$$d\vec{I} = A(v, w; \varepsilon) \vec{I}, \quad A = A_v dv + A_w dw, \quad (2)$$

with rational matrices — a *Pfaffian system*, in the terminology of the literature — subject to the integrability (flatness) condition $dA = A \wedge A$, verified exactly. The families themselves are already fixed by the reduction: propagator sets are identified modulo *affine* redefinitions of the loop

momenta, a linear map composed with a shift, $k_\ell \mapsto \sum_m M_{\ell m} k_m + \sum_r c_{\ell r} p_r$ with $M \in \text{GL}(L, \mathbb{Z})$, hence $|\det M| = 1$ and the measure invariant, that map cut lines to cut lines. The Symanzik polynomials (U, F) depend only on the graph and on the assignment of the external momenta to its vertices, not on the routing of the loop momenta, so equivalent families share them up to a permutation of the Feynman parameters; a coincidence of their normal orderings (Pak's algorithm) nominates a candidate, and the explicit momentum shift, constructed from the linear conditions $\tilde{D}_i(Mk + c \cdot p) = D_{\sigma(i)}(k)$ and applied, proves it. Because differentiation never leaves a family, the 347 masters split into 91 separate Pfaffian systems, of dimensions between 2 and 47 (the dimensions sum to more than 347 because each system carries the subsector masters it couples to, counted once in the family that owns them). The systems are separate but not independent: a subsector master shared by two of them is one and the same function, and they are solved in an order compatible with the subsector relation. Ordered by sectors — a sector is the set of propagators actually present — each connection is block lower triangular: diagonal blocks couple the masters of one sector group, and the entries below the diagonal feed subsector solutions upward.

2 Equivalence of blocks

Form the directed graph on the masters of a system with an edge $i \rightarrow j$ whenever I_i occurs in the derivative of I_j . A *block* is a strongly connected component of that graph, equivalently a diagonal block of the triangular structure. The 91 systems contain 1119 blocks. Two blocks with connections $b^{(1)}, b^{(2)}$ are *equivalent* when there is a constant permutation matrix P with

$$b^{(2)}(v, w) = P^{-1} b^{(1)}(v, w) P \quad \text{or} \quad b^{(2)}(v, w) = P^{-1} b^{(1)}(w, v) P, \quad (3)$$

the second option being the exchange $p_1 \leftrightarrow p_2$ of the incoming partons — an exact matrix identity, never an analogy. Since P is constant it carries no derivative term, so the equivalence is decided entry by entry. Only 173 equivalence classes occur, because the same subtopologies recur inside many parent families. The analysis is therefore done once for each class; a family is recovered afterwards from the classes it contains, and the identification is checked again at that point.

3 The canonical form

A system is in *canonical form* when

$$d\vec{F} = \varepsilon \sum_a R_a d \log \varphi_a \vec{F}, \quad (4)$$

with constant residue matrices R_a and ε -free *letters* φ_a (the *alphabet*). For two blocks in this form the equivalence of the preceding section reduces to a statement about the residues alone, $R_a^{(2)} = P^{-1} R_{\sigma(a)}^{(1)} P$, with σ the relabeling of the letters induced by the exchange. The Laurent expansion decouples order by order, $d\vec{F}^{(k)} = (\sum_a R_a d \log \varphi_a) \vec{F}^{(k-1)}$: each order is a quadrature of the previous one, and the solution is a path-ordered exponential whose order- ε^k term is a $\mathbb{Q}$ -linear combination of k -fold iterated integrals over the alphabet, contracted with the boundary vector — of uniform transcendental weight k , and, on segments where every letter is linear in the path parameter, generalized polylogarithms. Take the alphabet $\{w, 1 - w\}$ with

$$R_w = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}, \quad R_{1-w} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \vec{F} \rightarrow (1, 0)^T \quad \text{at } w \rightarrow 0. \quad (5)$$

The first component stays constant, and the recursion gives $F_2^{(1)} = \int_0^w d \log(1 - w') = \log(1 - w)$ and then $F_2^{(2)} = - \int_0^w d \log w' \log(1 - w') = \text{Li}_2(w)$: the dilogarithm as the twofold iterated integral of the word $(w, 1-w)$, and nothing more elaborate is involved. All singular behaviour is manifest: in the finite plane the connection has simple poles exactly at the letters' zeros, and the residue matrix R there prescribes the local solutions $H(w) w^{\varepsilon R} \vec{c}$ with H analytic and $H(0) = 1$, that is modes $w^{n+a\varepsilon} (\log w)^j$ — exponents from ε times the eigenvalues of R , logarithms at fixed ε from its Jordan structure — known before any integration. Two provisos: the normal form requires that no two exponents differ by a nonzero integer, automatic for generic ε ; and after expansion in ε logarithms appear at every order in any case, since $w^{\varepsilon R} = \sum_m (\varepsilon \log w)^m R^m / m!$, whether or not R is diagonalizable.

4 Construction of the transformation

Under a change of basis $\vec{I} = T \vec{F}$ the connection transforms as $A \mapsto T^{-1} A T - T^{-1} dT$, and the triangular structure decides the order in which the blocks are treated. A diagonal block never feels what lies below it, so each block is brought to canonical form on its own, once per class. Either one posits a rational T of bounded degree, over the alphabet supplied by the irreducible factors of the denominators of A and of the discriminants in its residue eigenvalues, and matches coefficients in $dT = A T - \varepsilon T \sum_a R_a d \log \varphi_a$ — a system bilinear in T and the residues, so one fixes candidate residues and solves linearly for T in each case; or one reduces higher poles by Moser's algorithm, which decides by a rank condition whether a rational transformation lowers the pole, and normalizes the residue eigenvalues by balances

$$T = \bar{P} + \frac{v - v_1}{v - v_2} P, \quad \bar{P} = 1 - P, \quad P = \frac{u \eta^\top}{\eta^\top u}, \quad (6)$$

with η a left eigenvector of the residue at v_1 , eigenvalue λ , and u a right eigenvector of the residue at v_2 , eigenvalue μ , and $\eta^\top u \neq 0$. Both conditions are needed: $P A_{(1)} \bar{P} = 0$ keeps the transformed connection free of a double pole at v_1 and $\bar{P} A_{(2)} P = 0$ does the same at v_2 , and a spectral projector of one residue satisfies only the first. The balance sends $\lambda \mapsto \lambda - 1$ and $\mu \mapsto \mu + 1$ and leaves the exponents at every other point unchanged, the residue at v_k being conjugated by the constant $T(v_k)$ and the one at infinity untouched since $T(\infty) = 1$; it must act at two points, since the sum of all residues including the one at infinity, $-\sum_k A_k$, vanishes identically, so no transformation changes an exponent at a single point. Repeated balances remove the integer parts n from all eigenvalues $n + a\varepsilon$ — each admissible step decreases the non-negative integer $\sum |n|$ by two, so the sequence cannot continue indefinitely; what can fail is the existence of an admissible pair with $\eta^\top u \neq 0$ at some step. A constant transformation then factors ε out of the residues. With two variables the whole construction is carried out in v with w a symbol, over $\mathbb{Q}(w, \varepsilon)$ — the singular points of that sliced equation are the v -roots of the letters, algebraic functions of w ; the freedom left is a v -independent $U(w; \varepsilon)$, which must both keep the conjugated residues $U^{-1} R_a U$ constant and satisfy the linear equation $\partial_w U = A'_w U - \varepsilon U \sum_a R_a \partial_w \log \varphi_a$ — a nullspace computation over $\mathbb{Q}(w, \varepsilon)$ whose solvability is decided by the solve itself and not by flatness, which is necessary and not sufficient. A candidate counts as the canonical form only when the transformed block equals $\varepsilon \sum_a R_a d \log \varphi_a$ identically in *both* variables. With the diagonal canonical, the entries below it yield to unipotent transformations $T = 1 + D$ (so $D^2 = 0$): writing εe_k and εc_j for the canonical diagonal blocks of row k and column j , and $\bar{b}_{kj}$ for the coupling to be removed (including the feed-through of previously completed entries of the same row), the unknown block D_{kj} and

the residues K_a of the completed entry — constant in the kinematics — satisfy the *affine-linear* equation

$$\mathrm{d}D_{kj} = \varepsilon(e_k D_{kj} - D_{kj} c_j) + \bar{b}_{kj} - \varepsilon \sum_a K_a \mathrm{d} \log \varphi_a. \quad (7)$$

A linear operator applied to the unknowns plus the fixed source $\bar{b}_{kj}$: its solution set is accordingly an affine subspace, one particular solution plus the null space of the linear part, the same structure the constraint system of the boundary analysis has below. It is solved by an ansatz over the letters with undetermined polynomial numerators, evaluated at many rational kinematic and regulator values modulo large primes; the exact rational solution — a rational function of ε as well — is reassembled by the Chinese remainder theorem and rational reconstruction, and proved by exact substitution. Where the solution is not unique, the representative with vanishing top-degree numerator coefficients is chosen: all are correct, their canonical forms differing by a constant transformation that preserves the diagonal blocks, and a low-degree one keeps the degrees of the later rows from compounding. What survives is a dependence on ε that is constant in the kinematics — a residue of the ε -dependent normalization of the basis — and it is removed last by a conjugation $T(\varepsilon)$, determined by the conditions $\tilde{R}_a(\varepsilon) T(\varepsilon) = \varepsilon T(\varepsilon) R_a$, linear in T , at a few rational kinematic points, and proved exactly on the full connection.

5 Square roots and their rationalization

Eliminating the direction of an unobserved momentum at fixed invariants requires solving quadratic equations, whose real solutions end where a Gram determinant vanishes; for three momenta $P_1, P_2, P_3 = -P_1 - P_2$ that determinant is a Källén polynomial,

$$\det \begin{pmatrix} 2P_1^2 & 2P_1 \cdot P_2 \\ 2P_1 \cdot P_2 & 2P_2^2 \end{pmatrix} = 4P_1^2 P_2^2 - (P_3^2 - P_1^2 - P_2^2)^2 = -\lambda(P_1^2, P_2^2, P_3^2), \quad (8)$$

with $\lambda(a, b, c) = (c - a - b)^2 - 4ab$ the triangle function. With the virtualities $s, -t, -u$ and λ homogeneous of degree two, this is, in units of s^2 , the Källén polynomial

$$\lambda_1 = \lambda(1, v, w) = (1 - v - w)^2 - 4vw = \left(1 - (\sqrt{v} + \sqrt{w})^2\right) \left(1 - (\sqrt{v} - \sqrt{w})^2\right), \quad (9)$$

whose second factor is positive on the whole triangle: the zero locus is $\sqrt{v} + \sqrt{w} = 1$, a pseudo-threshold *inside* the physical region, through $v = w = \frac{1}{4}$. Alphabets of many blocks therefore contain $\sqrt{\lambda_1}$ -type letters. Six radicands occur in all; the involution $v \leftrightarrow w$ exchanges two pairs of them and fixes the other two. The *root content* of a system is the number of independent square roots in its alphabet, counted so that $\sqrt{q_1}, \sqrt{q_2}, \sqrt{q_1 q_2}$ are two roots and not three — formally, the rank of the subgroup of $\mathbb{Q}(v, w)^\times / (\mathbb{Q}(v, w)^\times)^2$ that the radicands generate, computed as the rank over $\mathbb{F}_2$ of the matrix of exponent parities of the irreducible factors. Reading the radicands off the derived connection, from the discriminants of the irreducible quadratic factors of its singular loci and of its residue eigenvalues, gives an upper bound on that rank and not its value, since an apparent singularity contributes a discriminant but no root. In this channel about half the families have no root, a third have one, a dozen have two, and three have three. A single root is removed by a *rationalizing substitution*, a rational change of variables whose Jacobian determinant is not identically zero and under which the radicand becomes a perfect square: for instance $v = xy$, $w = (1 - x)(1 - y)$ gives $\lambda_1 = (x - y)^2$, so $\sqrt{\lambda_1} = x - y$ and the alphabet becomes rational in (x, y) . Such substitutions are the classical parametrization of a conic by the pencil of lines through

a rational point on it, here the point at infinity: writing $q = \alpha^2 w^2 + b w + c$ as a quadratic with square leading coefficient and setting $\sqrt{q} = \alpha w + \tau$, the squares cancel and leave the linear equation

$$w = \frac{c - \tau^2}{2\alpha\tau - b}, \quad \sqrt{q} = \alpha w + \tau, \quad (10)$$

so that (v, τ) is a rationalizing chart. Two roots are removed by applying the same step to the second radicand *in the variables produced by the first*, which requires that it still be of degree two there — the pullback of a conic under a degree-two map is in general a quartic — and that its leading coefficient be a square; for the Källén pairs both hold.

For three roots no rationalizing substitution exists. Such a substitution would be a dominant rational map from the plane onto the $(\mathbb{Z}/2)^3$ cover of the plane branched over the three conics, a complete intersection of three quadrics $r_i^2 = Q_i(v, w, h) = h^2 q_i(v/h, w/h)$ in $\mathbb{P}^5$, of degree eight like the cover. Its pairwise branch intersections are transverse except for finitely many simple tangencies, each lifting to two ordinary double points (nodes, locally $xy = z^2$); how many is immaterial, since nodes are rational double points whose resolution is crepant — the canonical class pulls back without correction — and adjunction leaves it trivial, $\mathcal{O}(6 - 6)$, with vanishing irregularity $h^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$ and geometric genus $p_g = 1$: a K3 surface. A unirational surface, one dominated by a rational map from the plane, is rational in characteristic zero by Castelnuovo's theorem, and a rational surface has $p_g = 0$. So one computes instead in the extension $\mathbb{Q}(v, w, \varepsilon)[r_1, r_2, r_3]/(r_i^2 - q_i)$, whose Galois group $(\mathbb{Z}/2)^3$ acts by sign changes. Already for one root the mechanism is complete: multiplication by $c_0 + c_1 r$ is the matrix $\begin{pmatrix} c_0 & c_1 q \\ c_1 & c_0 \end{pmatrix}$, differentiation obeys $dr = \frac{1}{2} r d \log q$, and a one-form equation in the algebra splits into two coupled equations with rational coefficients for the even and odd components. For three roots there are eight graded components; the eight sign embeddings must and do recombine consistently.

6 Paths and polylogarithms

With the canonical form established, evaluation means choosing a base point in the physical triangle and a path composed of segments on which every letter is linear in the path parameter. The criterion is what the generalized polylogarithms

$$G(a_1, \dots, a_k; t) = \int_0^t \frac{dt'}{t' - a_1} G(a_2, \dots, a_k; t'), \quad G(; t) = 1, \quad (11)$$

integrated along a path from 0 to t avoiding the a_i with a fixed branch of each logarithm, and completed for the words in which every a_i vanishes — the only ones the recursion cannot reach — by $G(\vec{0}_k; t) = \log^k t / k!$, require: if $\varphi_a(\gamma(t)) = \kappa_a(t - \alpha_a)$ then $d \log \varphi_a = dt / (t - \alpha_a)$, the constant dropping out, and every iterated integral along the segment is a G with indices the roots α_a . (Thus $G(1; t) = \log(1 - t)$ and $G(0, 1; t) = -\text{Li}_2(t)$, the two functions of the example above.) A letter quadratic on a segment stays quadratic on every sub-segment, so one either chooses the segments differently — in practice parallel to the axes, where the bilinear letters of the rationalizing substitutions become linear again — or admits the quadratic's linear factors as letters with algebraic constants. Flatness makes the result depend only on the endpoint and the homotopy class of the path; letters' zeros are crossed only as deliberate analytic continuation. The order in ε carried for each master is fixed in advance: if the assembled cross section is $\sigma = \sum_m c_m(\varepsilon) I_m$, with c_m carrying a pole of order p_m , and σ is wanted through ε^n , then I_m is wanted through ε^{n+p_m} . Of the 347 masters, 203 are needed only to their finite order and 131 one order deeper, 9 deeper still, and 4 less deeply.

7 Boundary conditions

The general solution leaves one constant vector per family per order, and almost all of it is fixed without evaluating any new integral. The most productive condition is the exclusion of local behaviour. Near $w = 0$ the equation offers

$$\vec{F}(v, w) = \sum_{\lambda} w^{\lambda} \vec{f}_{\lambda}(v, \log w), \quad \lambda \in \varepsilon \operatorname{spec} R_0 + \mathbb{Z}_{\geq 0}, \quad (12)$$

each $\vec{f}_{\lambda}$ linear in the boundary constants, while the integral representation admits only exponents $n + a\varepsilon$ produced by its momentum regions. The two lists must be compared in one basis: the regions constrain the master $\vec{I} = T\vec{F}$, and its exponents are those of the canonical $\vec{F}$ shifted by the order of T at the edge — which is also where the negative integer parts used in the next section come from. The regulator enters a region only through the measure, $d^d k \sim w^{4-2\varepsilon}$ for a soft momentum $k \sim w$ and $d^d k \sim w^{2-\varepsilon}$ for a collinear one, so $a = -2j_{\text{soft}} - j_{\text{coll}}$ counts the momenta a region scales. With two loop momenta, $j_{\text{soft}} + j_{\text{coll}} \leq 2$ and hence $a \in \{0, -1, -2, -3, -4\}$: five values, which is the whole spectrum observed on the soft edge. Every offered but forbidden mode gives $\vec{f}_{\lambda} \equiv 0$: one exact linear condition per mode — and the argument is the comparison of exponent lists, not an appeal to finiteness, since $w^{-\varepsilon}$ is itself bounded as $w \rightarrow 0$ for $\operatorname{Re} \varepsilon < 0$. The rest follows from elementary normalizations (the phase-space volume is a closed ratio of Γ -functions and fixes the lowest sector wherever it appears), from inheritance (families are treated in subsector order, and every already-known subsector master imposes its expansion wherever it recurs), and from regularity at letters' zeros interior to the region of convergence. All conditions assemble into one linear system $C\vec{C}^{(k)} = \vec{d}^{(k)}$ per order, so the number of genuinely undetermined constants is $\dim \ker C$, known before any evaluation. After identification across blocks — the same subsector being the same integral — at most thirty-three candidate *boundary periods* remain for the whole channel, values of cut phase-space integrals at ordered limits, ζ -values and kin , established by exact derivations with numerics as corroboration only. Their character is already visible in the beta integral $\int_0^1 dx x^{-1+a\varepsilon}(1-x)^{-1+b\varepsilon} = \Gamma(a\varepsilon)\Gamma(b\varepsilon)/\Gamma((a+b)\varepsilon)$, whose expansion is $\frac{a+b}{ab\varepsilon} [1 - ab\zeta_2\varepsilon^2 + O(\varepsilon^3)]$: a ζ -value out of Γ -functions alone. Where the parametric integrals do not factor, the Mellin–Barnes splitting $(X+Y)^{-\lambda} = \frac{1}{\Gamma(\lambda)} \frac{1}{2\pi i} \int dz \Gamma(\lambda+z)\Gamma(-z)X^zY^{-\lambda-z}$ separates them again.

8 Edges of phase space

Let u be the distance to an edge of the triangle, rescaled to $u \in [0, 1]$: $u = 1 - v - w$ on the soft edge, $u = v$ or $u = w$ on the collinear ones. The local solutions there are $u^{n+a\varepsilon}(\log u)^j$, with all data read from the residue matrix; a mode coefficient is extracted from the global solution as $\lim_{u \rightarrow 0} \Pi_{\lambda} u^{-\varepsilon R_u} \vec{F}$, the projection onto the generalized eigenspace of exponent λ , and on the soft edge it is a function of the variable surviving along that edge, not a number. The cross section then requires the distributional identity

$$u^{-1+a\varepsilon} = \frac{1}{a\varepsilon} \delta(u) + \sum_{k \geq 0} \frac{(a\varepsilon)^k}{k!} \left[\frac{\log^k u}{u} \right]_+, \quad \int_0^1 du [g]_+ f = \int_0^1 du g [f(u) - f(0)]. \quad (13)$$

It is proved in one line: integrating against f and adding and subtracting $f(0)$ gives the term $f(0) \int_0^1 u^{-1+a\varepsilon} du$, equal to $f(0)/(a\varepsilon)$, plus a remainder in which $f(u) - f(0)$ vanishes at the endpoint, so that the expansion in ε may be carried out term by term. Modes $u^{n+a\varepsilon}$ with $n \leq -2$

are treated by subtracting further Taylor terms and produce derivatives of $\delta(u)$; for $n = -2$ the subtraction is $f(u) - f(0) - uf'(0)$. As written the identity is exact on the collinear edges, where u runs over $(0, 1)$; on the soft edge the rescaling is by $1 - v$, so the δ coefficient carries $(1 - v)^{a\varepsilon}$. The identity exists only for the unexpanded power: expanding the master in ε first leaves $u^{-1} \log^k u$, non-integrable at the endpoint, and destroys the δ -content. The masters therefore carry their exact local mode data alongside the polylogarithmic expansion, and the identity is applied before the final expansion in ε .

9 Search and proof

Nothing depends on how a transformation was found: an ansatz, a fit, arithmetic modulo a prime produce candidates, and what makes a candidate a theorem is an exact identity — flatness for a connection, a numerics-free derivation for a boundary period, and for a canonical form

$$T^{-1}AT - T^{-1}dT = \varepsilon \sum_a R_a d \log \varphi_a, \quad (14)$$

verified as an identity of rational (or algebraic) matrices. Where a verification is done modulo a prime at random points, the bound is stated: a nonzero polynomial of total degree D vanishes at a uniform point of $\mathbb{F}_p^n$ with probability at most D/p , so a false identity survives r independent evaluations with probability at most $(D/p)^r$ — below 10^{-29} already at $r = 2$ for $D \leq 10^4$ and a sixty-three-bit prime. At the time of writing 88 of the 91 families carry canonical forms established in this way; the three three-root families await the Galois-algebraic treatment, and the higher boundary periods and the local mode analysis of every master at every edge remain open. The method is settled: each step is defined and each of its claims is decided by an exact identity. What remains is its execution on what is still open.